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Evaluation report concerning the thesis of Jayani Lakshika Piyadi Gamage
by Anne M. Ruiz

The thesis by Jayani Lakshika Piyadi Gamage studies the problem of visualizing high-dimensional data using nonlinear dimension reduction methods (NLDR) such as t-SNE and UMAP. These approaches are now widely used because they often produce clear low-dimensional representations, with visible clusters or structures. The dissertation focuses on two-dimensional representations.

The main point of the thesis is that these visualizations can be misleading, since the methods may distort the original geometry of the data and even create structures that are not really present.

The dissertation is organized into eight chapters.

The first chapter provides a brief introduction that gives a short presentation of the objectives and the structure of the work.

In my opinion, Chapter 2 is the most important contribution of the thesis. It corresponds to a paper currently under revision for the very good “Journal of Graphical and Computational Statistics”, and focuses on the assessment and the interpretation of the NLDR embeddings. The key idea is to analyze the link between the low-dimensional representation and the original high-dimensional space, instead of only looking at the final plot. For this, the author introduces a diagnostic approach based on tour visualization, which makes it possible to explore how the embedding reflects the full data structure. The method is both clever and relatively sophisticated, even though it remains easy to use in practice.

Chapter 3 introduces an R package that implements the methodology developed in the previous chapter and provides practical tools for analyzing two-dimensional NLDR embeddings

Chapter 4 presents another R package dedicated to the generation of simulated datasets, enabling the evaluation of dimension reduction techniques in controlled settings.

Chapter 5 contains an original contribution based on an experimental study of how users interpret visualizations, showing that patterns such as clusters are often over-interpreted. This chapter is also particularly interesting as it highlights the candidate’s ability to design and conduct an experiment involving more than one hundred participants.

An interactive (Shiny) application is proposed in Chapter 6 designed to compare different embeddings and assist in their evaluation.

Chapter 7 concludes the manuscript and gives perspectives, while Chapter 8 provides details on the reproducibility and the availability of the material.

The manuscript also includes several appendices that are helpful for understanding the methodology in greater depth. In particular, Appendix A9 discusses some specific behaviors of NLDR methods, such as the apparent ordering of points or the compression of low-density regions in the layout. A detailed glossary is also provided.

Overall, the thesis combines important and interesting methodological developments together with practical tools to better understand and assess nonlinear dimension reduction methods. The dissertation is very pleasant to read because it gives a wide range of tools and studies for evaluating NLDR layouts that are very well exposed. The functionalities of the many R packages (3) are also very well explained and helpful for the user. It is evident that Jayani Lakshika Piyadi Gamage has a solid expertise in her research area, strong skills in developing sophisticated R packages, and is committed to disseminating the methods she develops.

The manuscript is very well structured and clearly written. All chapters are of interest. The objectives are clear and the methods proposed are sound and original. Numerous artificial and real data sets are analysed, providing convincing illustrations of the approach.

To conclude, my opinion on Jayani Lakshika Piyadi Gamage's thesis is very positive. The manuscript fulfills completely the requirements of an academic thesis for the award of a PhD.



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